

- Density (ρ) = $\frac{\text{mass (m)}}{\text{volume (v)}}$

- unit = kg/m^3

- density of water = $1000 \text{ kg / m}^3 = 1 \text{ g/cc}$

- for two bodies of same mass

$$\rho_1 V_1 = \rho_2 V_2$$

if $\rho_1 > \rho_2$

$$V_1 < V_2$$

Mixing of liquid

Calculation of resultant/final density

1) If volumes of the liquids are equal

$$d = \frac{d_1 + d_2}{2}$$

2) If masses of two liquids are equal

$$\text{For 2- liquids} \Rightarrow d = \frac{2d_1 d_2}{d_1 + d_2}$$

$$\text{For 3- liquids} \Rightarrow d = \frac{3d_1 d_2 d_3}{d_1 d_2 + d_2 d_3 + d_1 d_3}$$

$$\text{For n-liquids, } n = 1 + 1 + 1 + \dots + 1$$

$$d = \frac{1}{\frac{1}{d_1} + \frac{1}{d_2} + \frac{1}{d_3} + \dots + \frac{1}{d_n}}$$

3) If masses and volumes of two liquids are different

$$d = \frac{\text{Total mass}}{\text{Total volume}} = \frac{m_1 + m_2}{V_1 + V_2}$$

$$m_1 = \rho_1 V_1 \text{ \& } m_2 = \rho_2 V_2$$

$$V_1 = \frac{m_1}{\rho_1} \text{ \& } V_2 = \frac{m_2}{\rho_2}$$

Relative density (R.D)/Specific gravity

1) Relative density of a body

$$(R.D)_{\text{solid}} = \frac{d_{\text{solid}}}{d_{\text{water}}} = \frac{w_a}{w_a - w_w}$$

2) Relative density of liquid

$$(R.D)_{\text{liquid}} = \frac{d_{\text{liquid}}}{d_{\text{water}}} = \frac{w_a - w_L}{w_a - w_w}$$

3) Relative density of a solid to that of liquid

$$\frac{(R.D)_{\text{solid}}}{(R.D)_{\text{liquid}}} = \frac{w_a}{w_a - w_L}$$

weight of object where,
 W_a = weight of object when in air
 W_w = weight of object when dipped in water
 W_L = weight of object when dipped in liquid

Pressure

Normal force or thrust exerted by liquid per unit area

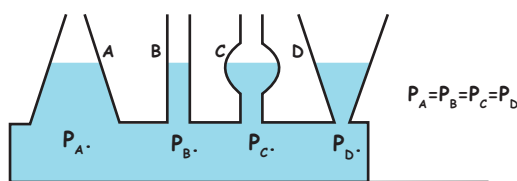
$$P = \frac{F}{A}$$

Pressure-depth relation

$$P = h \rho g$$

Hydrostatic paradox

Whatever the shape or width of vessel the pressure at any particular depth is same



- Gauge pressure = $P - P_{\text{atm}} = h \rho g$

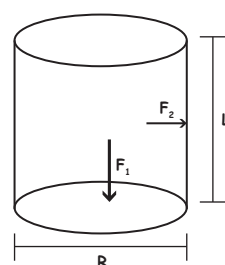
$$P_{\text{atm}} = 1.01325 \times 10^5 \text{ Pa}$$

Bubble rising up at constant temperature

radius 'r' becomes 'nr' when bubble rises in liquid from bottom to the surface

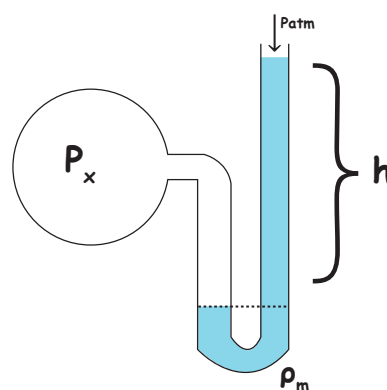
$$\rho g h = p_{\text{atm}} [n^3 - 1]$$

Conditions for equal forces on wall and bottom of a cylinder



If $L = R$
 then, $F_1 = F_2$

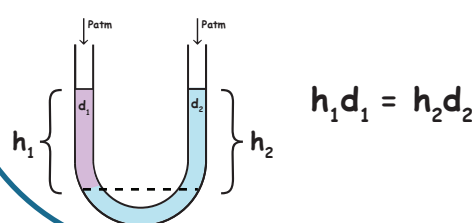
Manometer



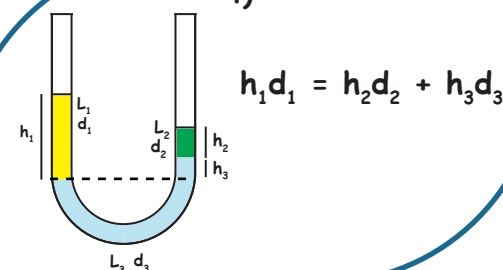
$$P_x = P_{\text{atm}} + h \rho_m g$$

$$h \rho_m g = P_x - P_{\text{atm}}$$

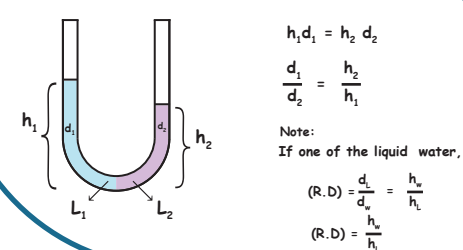
1) U-Tube manometer



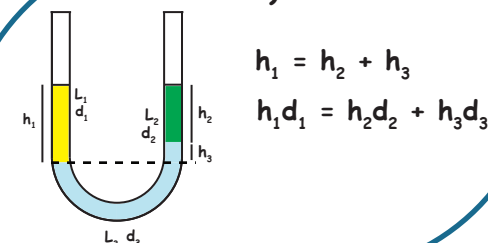
4)



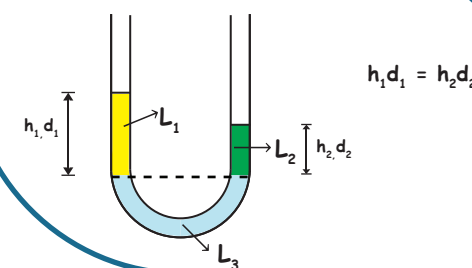
2) U-Tube type



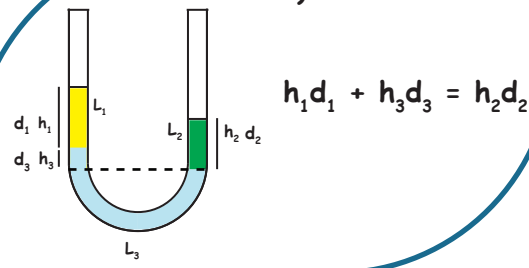
5)



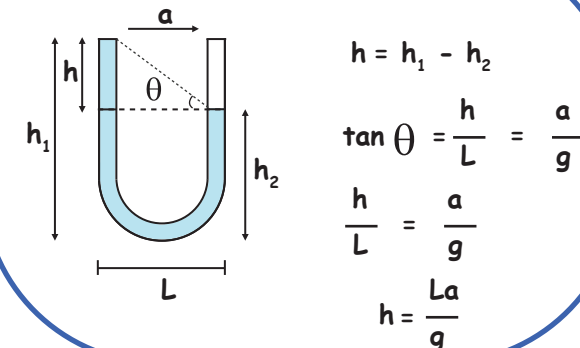
3) The third liquid is in level with other



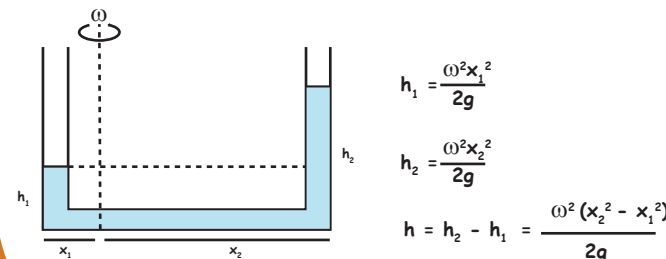
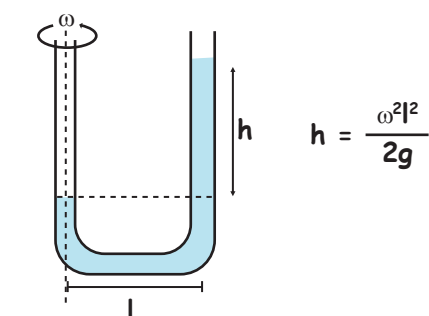
6)



U - tube accelerating horizontally



Special case: U - tube rotating



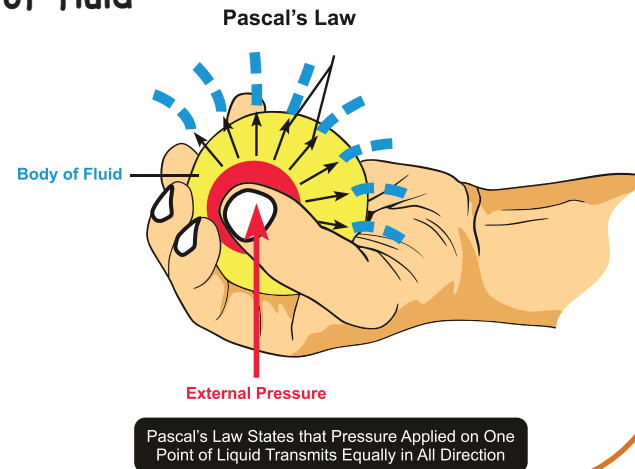
FLUID MECHANICS 01



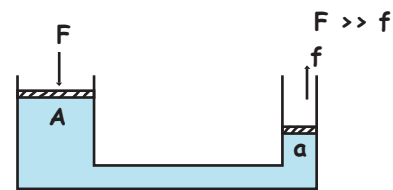
PHYSICS WALLAH

Pascal's Law

Any change in pressure at a point of an enclosed incompressible fluid is equally transmitted at all other points of fluid



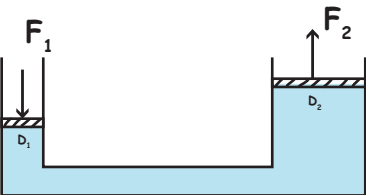
Application Hydraulic Lift



$$\frac{F}{A} = \frac{f}{a}$$

As $A \gg a$ therefore

If the cylinders are connected

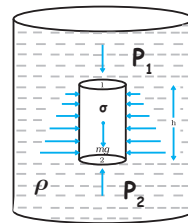


$$\frac{F_1}{\pi R_1^2} = \frac{F_2}{\pi R_2^2}$$

$$\frac{F_1}{R_1^2} = \frac{F_2}{R_2^2}$$

$$\frac{F_1}{D_1^2} = \frac{F_2}{D_2^2}$$

Archimede's principle



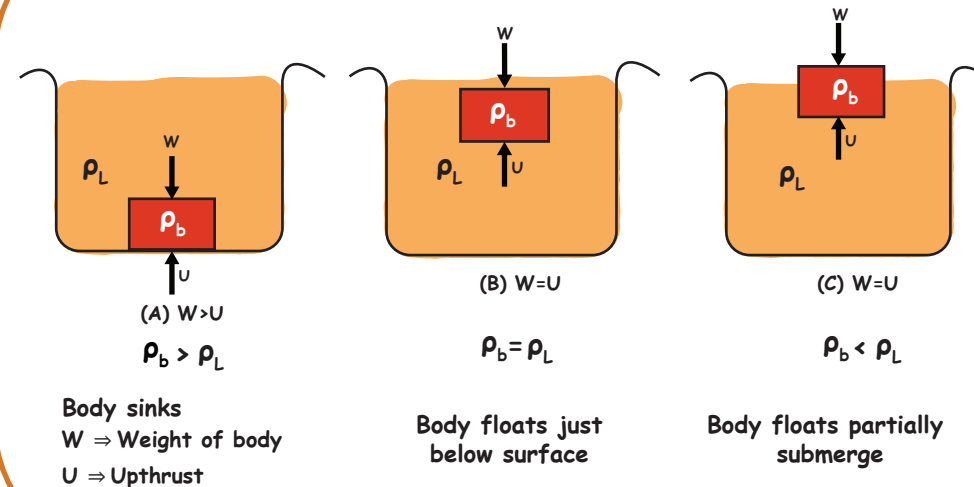
liquid applies net upward force on an immersed body, called as upthrust or buoyant force
upthrust = weight of the liquid displaced = $V\rho g$
Apparent weight = Actual weight - upthrust

$$W_{app} = W_{air} - U$$

$$= W_{air} \left[1 - \frac{\sigma}{\rho} \right]$$

$$P_2 > P_1, P_2 - P_1 = \text{Upthrust}(U)$$

Law of floatation



Body sinks
 $W \Rightarrow$ Weight of body
 $U \Rightarrow$ Upthrust

Body floats just below surface

Body floats partially submerge

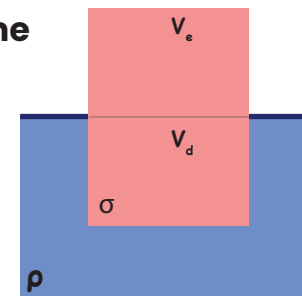
Fractional submerged volume

$$\frac{\text{Displaced volume}(V_d)}{\text{Total volume}(V)} = \frac{\sigma}{\rho} \text{ (submerged fraction)}$$

$$\frac{\text{Exposed volume}(V_e)}{\text{Total volume}(V)} = 1 - \frac{\sigma}{\rho} \text{ (Exposed fraction)}$$

$$\text{Relative density of a solid} = \frac{\text{weight of solid in air}}{\text{Loss of weight in water}} = \frac{w_a}{w_a - w_w} = \frac{\rho_b}{\rho_w}$$

$$\text{Relative density of a liquid} = \frac{\text{Loss of weight in liquid}}{\text{Loss of weight of an object dipped in water}} = \frac{w_a - w_l}{w_a - w_w} = \frac{\rho_l}{\rho_w}$$



Newton's Law of viscosity

$$F \propto A \frac{dv}{dx} \Rightarrow F = \eta A \frac{dv}{dx}$$

Where, Velocity gradient = $\frac{dv}{dx}$

$$\eta = \frac{F}{A \frac{dv}{dx}} \Rightarrow \text{coefficient of viscosity} \Rightarrow \eta = \frac{F/A}{\frac{dv}{dx}} = \frac{F/A}{\frac{v/l}{\left(\frac{dx}{dt}\right)/l}} = \frac{F/A}{\frac{d}{dt} \left(\frac{x}{l} \right)}$$

$$\Rightarrow \eta = \frac{\text{shearing stress}}{\text{strain rate}}$$

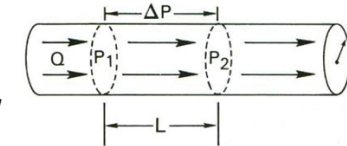
Unit of Coefficient of viscosity

- 1) The CGS Unit of η is dyne s cm⁻² and is called poise.
 - 2) The SI unit of η is Nsm⁻² called decapoise or poiseuille
- 1 poiseuille = 10 poise

Poiseuille's formula

$$Q = \frac{\pi \Delta P r^4}{8 \eta L}$$

where, Q = rate of flow



Stoke's law

$$F = 6\eta\pi r v$$

F_{net} = Apparent weight - viscous force

Terminal velocity

$$V_t = \frac{2r^2}{9\eta} (\rho - \sigma) g$$

- 1) If $\rho > \sigma$, the body will attain terminal velocity in the downward direction.
- 2) If $\rho < \sigma$ the terminal velocity will be negative and the body will move in the upward direction.
- 3) $\rho = \sigma$, the body remain suspended in the fluid.

Critical velocity

Reynold's number

$$R_e = \frac{\rho v D}{\eta}$$

Significance of Reynold's number:

- If R_e lies between 0 and 2000 the flow is streamlined or laminar.
- If $R_e > 3000$, the flow is turbulent.
- If R_e lies between 2000 & 3000 the flow of liquid is unstable. It may change from laminar to turbulent and vice versa.

Equation of continuity

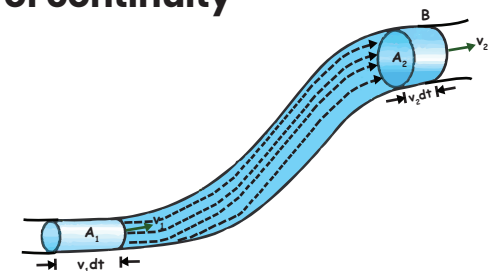
$$v_1 A_1 \Delta t \rho_1 = v_2 A_2 \Delta t \rho_2$$

since the liquid is incompressible $\rho_1 = \rho_2$

$$v_1 A_1 = v_2 A_2$$

$Av = \text{constant}$.

$$Av = \frac{dV}{dt} = Q \Rightarrow \text{Volume rate of flow}$$



FLUID MECHANICS 02

Energy of fluid in a study flow

$$\text{kinetic Energy} = \frac{1}{2} mv^2$$

$$\text{kinetic energy per unit mass} = \frac{1}{2} v^2$$

$$\text{kinetic energy per unit volume} = \frac{1}{2} \rho v^2$$

$$\text{Potential Energy} = mgh$$

$$\text{Potential energy per unit mass} = gh$$

$$\text{Potential energy per unit volume} = \rho gh$$

$$\text{Pressure energy} = PV$$

$$\text{Pressure energy per unit mass} = \frac{P}{\rho}$$

$$\text{Pressure energy per unit volume} = P$$

BERNOULLI'S PRINCIPLE

$$P_1 V_1 - P_2 V_2 = \frac{1}{2} m (v_2^2 - v_1^2) + mg (h_2 - h_1)$$

$$(P_1 - P_2) V = \frac{1}{2} m (v_2^2 - v_1^2) + mg (h_2 - h_1)$$

$$(P_1 - P_2) V = \frac{1}{2} \frac{m}{V} (v_2^2 - v_1^2) + \frac{mg}{V} (h_2 - h_1)$$

$$\Rightarrow P + \frac{1}{2} \rho v^2 + \rho gh = \text{constant}$$

$$\frac{P}{\rho g} + \frac{v^2}{2g} + h = \text{constant}, \quad \frac{P}{\rho g} = \text{pressure head}$$

$$\frac{v^2}{2g} = \text{velocity head} \quad h = \text{Gravitational head}$$

Conditions:

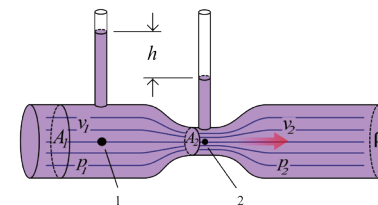
1. Flow should be streamlined.
2. Non-viscous and incompressible fluid.
3. Friction is absent everywhere.

Note: It is based on conservation of energy.

VENTURIMETER

Device to measure the flow of speed of incompressible fluid

$$v_1 = \sqrt{\frac{2hg}{(A_1^2/A_2^2) - 1}}$$

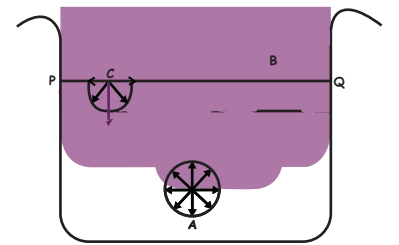


SURFACE TENSION

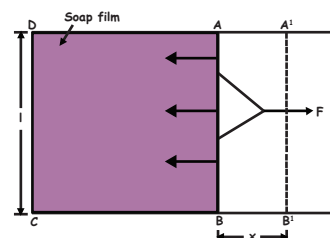
$$\text{Surface tension } T = \frac{\text{Force}}{\text{Length}} = \frac{F}{l}$$

$$\text{Unit in SI system} = \frac{\text{N}}{\text{m}}$$

$$\text{Unit in CGS system} = \text{dyne / cm}$$



SURFACE ENERGY



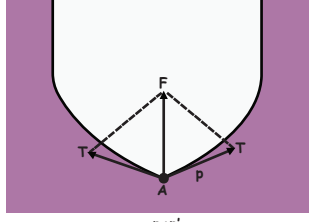
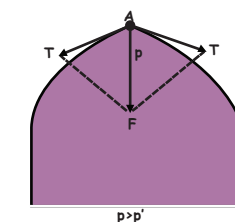
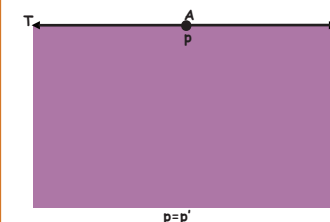
$$\text{Work done } W = F \times x$$

$$\text{But } F = 2TL$$

$$\Rightarrow W = 2TL \times x$$

$$\text{Energy of the additional surface} = W = 2TLx \\ = T(2Lx) = T\Delta A$$

PRESSURE DIFFERENCE ACROSS A CURVED LIQUID SURFACE

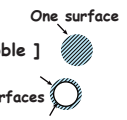


Pressure on concave side > pressure on convex side

$$P_{\text{concave}} - P_{\text{convex}} = \frac{2T}{R}$$

$$P_{\text{inside}} - P_{\text{outside}} = \frac{2T}{R} \quad [\text{Liquid drop or air bubble}]$$

$$P_{\text{inside}} - P_{\text{outside}} = \frac{4T}{R} \quad [\text{Soap bubble}]$$



APPLICATIONS OF BERNOULLI'S PRINCIPLE

Torricelli's Law of Efflux

$$v = \sqrt{\frac{2(P - P_a)}{\rho} + 2gh}$$

- It is assumed that size of hole << size of top of tank

If tank is open, $P = P_a$

$$\text{Then } v = \sqrt{2gh}$$

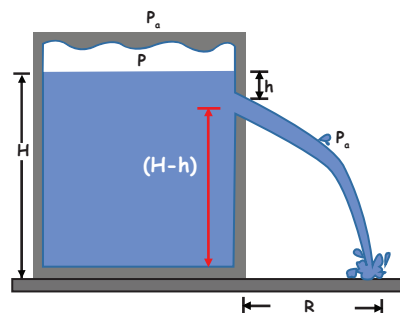
$$\text{Time of fall, } t = \sqrt{\frac{2(H-h)}{g}}$$

$$\text{Range } R = vt$$

$$= \sqrt{2gh} \times \sqrt{2(H-h)/g}$$

$$\Rightarrow R = 2\sqrt{h(H-h)}$$

$$R \text{ is max, when } h = \frac{H}{2}$$



Excess pressure inside a liquid drop

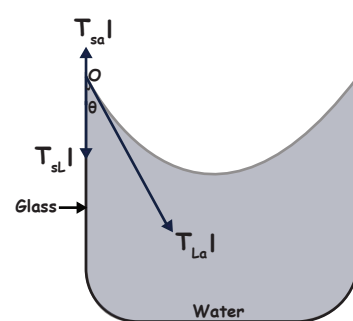
$$P_i - P_o = \frac{2T}{R}$$

Excess pressure inside a soap bubble

$$P_i - P_o = \frac{4T}{R}$$

FLUID MECHANICS 03

Shape of liquid meniscus



Consider the equilibrium along the surface at line of contact

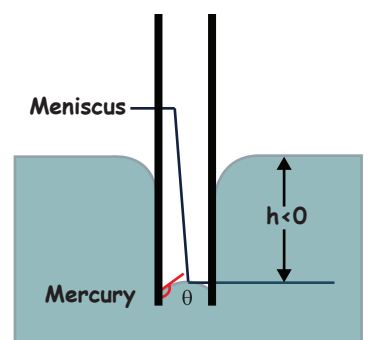
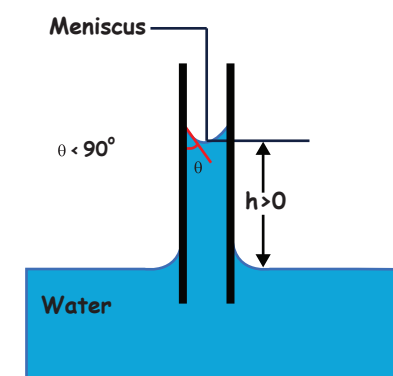
$$T_{sa}l = T_{sl}l + T_{la}l \cos \theta$$

$$T_{sa} = T_{sl} + T_{la} \cos \theta$$

$$\cos \theta = \frac{T_{sa} - T_{sl}}{T_{la}}$$

θ = Angle of contact.

Capillarity



Ascent/descent formula:

$$h = \frac{2T}{R\rho g}, \quad h > 0 (\theta < 90^\circ) \quad h = \frac{2T \cos \theta}{r\rho g}, \quad h < 0 (\theta > 90^\circ)$$

where, R = radius of meniscus
 r = radius of the tube